# Does the counterterm $\alpha_2$ double count the Fingers-of-God dispersions?

The redshift-space counterterm against the sampled velocity dispersions $\sigma_v$, template by template and in the posterior

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## 1 Summary

**The question.** The forecast's one-loop model carries two descriptions of nonlinear velocities along the line of sight. One is the Fingers-of-God blur: each bin's redshift distribution is convolved with a Gaussian whose width is set by a velocity dispersion $\sigma_v$, one per dispersion group, sampled under a Gaussian prior. The other is the line-of-sight counterterm $\alpha_2$ of the DESI 2024 full-shape convention [1],

$$P_{\rm CT}(k,\mu) \;=\; \left(b_1 + f\mu^2\right) f\,\alpha_2\,\mu^2 k^2\,P_{\rm lin}(k), \tag{1}$$

with $k$ the comoving wavenumber, $\mu$ the cosine of the angle to the line of sight, $b_1$ the bin's linear galaxy bias, $f = d\ln D/d\ln a$ the linear growth rate ($D$ the linear growth factor, $a$ the scale factor), $P_{\rm lin}$ the linear matter power spectrum, and $\alpha_2$ in $h^{-2}\,{\rm Mpc}^2$, one value per group plus a free tilt in redshift in model B + tilts. The leading effect of the blur on the power spectrum is a damping $-\sigma_\chi^2 k_\parallel^2$ with $k_\parallel = k\mu$ and $\sigma_\chi = (1+z)\,\sigma_v/H(z)$ the dispersion in comoving distance: the same $k_\parallel^2$ shape as Eq. (1) (Section 2). So the question is whether the forecast pays twice for one effect: is $\alpha_2$ degenerate with the $\sigma_v$ already in the model?

**The answer.**

1. **At the template level, yes.** On every layout and at every cut, combinations of the seven $\sigma_v$ templates reproduce 95.8–100% of each $\alpha_2$ group template's information in the $\mathsf{C}^{-1}$ metric ($\mathsf{C}$ the band-power covariance), with medians 99.3–99.9%; the two templates are mirror images, so a positive $\alpha_2$ looks like a smaller $\sigma_v$ (Figure 1). Only the $\alpha_2$ tilt columns are substantially new.

2. **In the posterior, the $\sigma_v$ priors break the degeneracy at small cuts but not at large ones.** In the paper model (model B + tilts on the pessimistic linear-bias prior) the largest correlation between an $\alpha_2$ parameter and a $\sigma_v$ grows, on the thin layouts, from $|\rho| = 0.39$–$0.46$ at $k_{\rm max} = 0.075\,h/{\rm Mpc}$ to 0.81–0.88 at 0.2 (Figure 2).

3. **What $\alpha_2$ costs.** Fixing $\alpha_2$ and its tilts at zero raises ${\rm FoM}(w_0, w_a)$ of the paper model by a factor of 2.4–2.7 at 0.075 and 1.16–1.25 at 0.2 on the thin layouts; ${\rm FoM}(\Omega_m, 10^9 A_s)$ moves the same way. The gain comes entirely from the $\sigma_v$ priors: with flat $\sigma_v$ priors, fixing $\alpha_2$ returns 0.83–1.01 of the paper model's FoM at every cut from 0.075 on.

4. **What dropping $\alpha_2$ risks.** If the truth holds a DESI-prior-sized $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group, a fit without $\alpha_2$ (model A + tilts) is shifted in $(w_0, w_a)$, and $\sigma_v$ on its priors absorbs almost none of it: the shift falls by 4% and 17% at 0.075 and by 35% and 41% at 0.2 on the two thin layouts. With flat $\sigma_v$ priors those shifts fall by 88–97%. The wide layouts barely react either way.

5. **A wider $\sigma_v$ prior trades the gain for the absorption.** Multiplying all seven $\sigma_v$ prior widths by a common factor $s$, the fit without $\alpha_2$ absorbs the $12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth (every shift below $\sigma/3$) at $s = 1.84$ and 1 at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ on the thin layouts, where it still keeps 2.1 and 2.8 times the paper model's FoM; at 0.1–0.2 it absorbs only at $s = 3.6$–$6.0$, where the FoM ratio is 0.96–1.22 and the gain is gone (Figure 3). Widening one sample's dispersions alone never suffices at 0.1 and above.

6. **Sampling $\sigma_v$ can outgrow the pair list.** The data vector's cross pairs are fixed by an overlap rule at the fiducial $\sigma_v$ ($O_{ij} \geq 10^{-6}$, Section 6). Chains over the current priors stay inside the list: the nearest unlisted pair reaches $7.6 \times 10^{-7}$ at the $+2\sigma$ edge on $\Delta d_{\mathrm{com}} = 62$ Mpc, which crosses the threshold at $+2.1$ prior widths. With the widened priors of item 5 the crossing sits at 0.5–0.8 of the widened width, and with flat $\sigma_v$ a DESI-size negative $\alpha_2$ pushes $\Delta z = 0.02$ across at $|\alpha_2| \approx 16$–$19\,h^{-2}\,\mathrm{Mpc}^2$.

Here $w_0$ and $w_a$ parameterize the dark-energy equation of state $w(a) = w_0 + w_a(1-a)$; $\Omega_m$ is the matter density today in units of the critical density; $A_s$ is the amplitude of the primordial curvature power spectrum, sampled as $10^9 A_s$; and $\mathrm{FoM}(w_0, w_a) = [\det \mathrm{Cov}(w_0, w_a)]^{-1/2}$ is the inverse area of the marginalized $(w_0, w_a)$ ellipse, likewise $\mathrm{FoM}(\Omega_m, 10^9 A_s)$. In short: one $k_\parallel^2$ effect, counted twice at the template level; the posterior keeps the two apart only as long as the $\sigma_v$ priors do, and the whole dark-energy gain of dropping $\alpha_2$ is exactly as strong as those priors. Whether to keep both, and on what basis for the $\sigma_v$ priors, is an open decision (appendix "Open issues").

**The setup.** The data are the DESI angular clustering band powers $C_\ell^{gg}$ alone, computed without the Limber approximation, on four bin layouts: $\Delta d_{\mathrm{com}} = 62$ Mpc (equal comoving width) and $\Delta z = 0.02$, 0.05 and 0.2 (fixed redshift width); the first two are the thin layouts. The scan runs over $k_{\max} = 0.05$ to $0.2\,h/\mathrm{Mpc}$, each the $z = 0$ value of a per-bin cut at $k_{\max}/D(\bar{z}_i)$, with $\bar{z}_i$ the mean redshift of bin $i$. Every number and figure of this note comes from the final section of `EXAMPLE_EVALUATE_FISHER_MODEL_B.ipynb`; the per-cut values are in the two figure folders' `data.json`.

## 2 Why the question is natural

**Redshift is blurred distance.** The forecast places each galaxy at its measured redshift. Inside groups and clusters, galaxies orbit with random speeds of several hundred km/s, so a galaxy's measured redshift carries a Doppler shift on top of its distance: in a redshift map a cluster stretches into a finger pointing at the observer, the Fingers of God [3]. A line-of-sight velocity $v$ at redshift $z$ displaces the galaxy by

$$\delta\chi \;=\; \frac{(1+z)\,v}{H(z)}, \tag{2}$$

with $\chi$ the comoving distance and $H(z)$ the expansion rate. A dispersion $\sigma_v$ of these velocities blurs distances by $\sigma_\chi = (1+z)\,\sigma_v/H(z)$: for the groups here, 5–6 Mpc, about $4\,h^{-1}$Mpc (`notes/cross_correlations`, which derives the convolution and its sizes).

**Blurring the window equals blurring the field.** The likelihood implements the blur by convolving each bin's redshift distribution $n_i(z)$ with a Gaussian of width $(1+\bar{z}_i)\,\sigma_v/c$ in redshift [5]. That window-side convolution is the same operation as damping the three-dimensional field, as long as the velocities are a random scatter uncorrelated with the density. Write the observed galaxy field as the true one with each galaxy moved by $\delta\chi$ drawn from a distribution $p$: along one line of sight, $\delta_g^{\rm obs}(\chi_{\rm obs}) = \int d\chi\, p(\chi_{\rm obs}-\chi)\,\delta_g(\chi)$. A bin's map integrates the observed field over its radial window $W_i$:

$$\begin{aligned}\int d\chi_{\rm obs}\, W_i(\chi_{\rm obs})\,\delta_g^{\rm obs}(\chi_{\rm obs}) \;&=\; \int d\chi \Big[\int d\chi_{\rm obs}\, W_i(\chi_{\rm obs})\, p(\chi_{\rm obs}-\chi)\Big]\,\delta_g(\chi)\\ &=\; \int d\chi\,\big(W_i{*}p\big)(\chi)\,\delta_g(\chi)\,: \end{aligned}\tag{3}$$

swapping the order of integration moves the convolution from the field onto the window. Blurring every bin's $n(z)$ and keeping the true field is therefore the same map as keeping sharp bins and blurring the field, which is how the likelihood's $\sigma_v$ reach the spectra.

**The damping and its leading order.** On the field side, a convolution along the line of sight multiplies each Fourier mode by the distribution's transform: for a Gaussian $p$ of width $\sigma_\chi$, the density picks up $\exp(-k_\parallel^2\sigma_\chi^2/2)$ and the power spectrum $\exp(-k_\parallel^2\sigma_\chi^2)$, with $k_\parallel = k\mu$ the line-of-sight wavenumber. Applied to the linear redshift-space (Kaiser [4]) power, to leading order in $k_\parallel^2\sigma_\chi^2$,

$$\big(b_1+f\mu^2\big)^2 P_{\rm lin}\, e^{-k_\parallel^2\sigma_\chi^2} \;\simeq\; \big(b_1+f\mu^2\big)^2 P_{\rm lin} \;-\; \sigma_\chi^2\, k_\parallel^2\,\big(b_1+f\mu^2\big)^2 P_{\rm lin}.\tag{4}$$

The counterterm, Eq. (1), is $f\,\alpha_2\,k_\parallel^2\,(b_1+f\mu^2)\,P_{\rm lin}$. Both are $k_\parallel^2$ times the linear power; their $\mu$ structures differ by exactly one factor of $(b_1+f\mu^2)$, and a band power averages much of that difference away. The signs are opposite where both amplitudes are positive: a positive $\alpha_2$ raises the spectrum where a larger $\sigma_v$ lowers it, so a positive $\alpha_2$ looks like a *smaller* $\sigma_v$. The sizes are comparable too: $\sigma_\chi \approx 4\,h^{-1}$Mpc gives $\sigma_\chi^2 \approx 16\,h^{-2}\,{\rm Mpc}^2$, the order of the DESI prior width $12.5\,h^{-2}\,{\rm Mpc}^2$ of $\alpha_2$ at $k_{\rm max} = 0.2\,h/{\rm Mpc}$.

**Why both exist.** In a three-dimensional full-shape analysis there is no radial window to convolve, so there is no sampled $\sigma_v$: the counterterm $\alpha_2$ is that analysis's effective description of small-scale velocities, Fingers of God included [1, 2]. This forecast works on projected spectra, where the window-side convolution is natural and $\sigma_v$ already carries the effect; $\alpha_2$ was added with the rest of the DESI counterterm set. A setup with $\alpha_0$ only – the density counterterm, which multiplies $k^2 P_{\rm mm}$ with no $\mu$ – is the natural alternative, and this note measures what each choice costs.

**The seven dispersions and their priors.** Each bin's $\sigma_v$ is its dispersion group's value (Table 1): the measured dispersion curve of the source forecast's simulated galaxies steps

inside BGS, so BGS takes two values, split at $\bar{z} = 0.255$, giving seven dispersion groups for six bias groups. Each prior is a Gaussian centred on the measured value, five times as wide as that measurement's per-bin error bar, and truncated at zero; the defaults and priors live in `likelihood/params_desi*.yaml` (the `DESI_SIGV` parameters, one sampled value per group), and their provenance is the section "Where the velocity dispersions come from" of `notes/cross_correlations`. The same seven values and widths serve every layout.

| dispersion group | $\sigma_v$ [km/s] | prior width [km/s] |
|---|---|---|
| BGS, $\bar{z}_i < 0.255$ | 371 | 145 |
| BGS, $\bar{z}_i \geq 0.255$ | 358 | 105 |
| LRG1 | 357 | 70 |
| LRG2 | 348 | 50 |
| LRG3 | 343 | 50 |
| ELG1 | 304 | 40 |
| ELG2 | 289 | 30 |

Table 1: The fiducial velocity dispersion of each group and the width $\Delta\sigma_{v,g}$ of its Gaussian prior.

## 3 Templates, and how to measure redundancy

**A parameter's template.** A parameter $p$'s template is the derivative of the data vector, $\partial C_\ell/\partial p$ over every kept band of every spectrum. To first order it is the fingerprint $p$ leaves on the data: the fit can distinguish two parameters only as far as their templates differ. Each parameter is measured in units of its prior width $\Delta_p$ (written $\Delta$, never $\sigma$, since $\sigma_v$ is itself a dispersion): here $\Delta\alpha_2 = 0.5/k_{\rm max}^2$, the model B prior width, quoted at $k_{\rm max} = 0.2\,h/{\rm Mpc}$ where it is $12.5\,h^{-2}\,{\rm Mpc}^2$, and $\Delta\sigma_{v,g}$ each dispersion group's prior width (Table 1). Divided by each band's error $\sigma_\ell = \sqrt{\mathsf{C}_{ii}}$, the scaled template $\sigma_\ell^{-1}\,\partial C_\ell/\partial(p/\Delta_p)$ reads directly as how many error bars the parameter can move each band while staying within its prior; Figure 1 draws the templates this way, following Fig. 1 of Obied et al. [7]. The $\alpha_2$ templates here are group templates: the response to the group's value, all bins of the group moving together (and, for the tilts, moving along a straight line in redshift).

**The leftover, measured in the data's own error bars.** Can an $\alpha_2$ group template be rebuilt as a weighted sum of the seven $\sigma_v$ templates? The leftover of the best weights $c_g$ is

$$r_\ell \;=\; \frac{\partial C_\ell}{\partial(\alpha_2/\Delta\alpha_2)} \;-\; \sum_g c_g\,\frac{\partial C_\ell}{\partial(\sigma_{v,g}/\Delta\sigma_{v,g})}, \tag{5}$$

the sum running over the seven dispersion groups. It should not be measured band by band: a mismatch of $0.1\sigma_\ell$ on a precise band is visible where the same mismatch on a noisy band is not. The natural norm is the data's own, $r^{\mathsf{T}}\mathsf{C}^{-1}r$ with $r$ the vector of the $r_\ell$, the $\chi^2$ the leftover would contribute, and the weights $c_g$ minimize it. Writing $a$ for the scaled $\alpha_2$ template, the first term of Eq. (5), the redundancy fraction is the share of the template's information that the $\sigma_v$ templates reproduce,

$$f \;=\; 1 \;-\; \frac{\min_c\, r^{\mathsf{T}}\,\mathsf{C}^{-1}\,r}{a^{\mathsf{T}}\,\mathsf{C}^{-1}\,a}. \tag{6}$$

$f = 1$ means the data cannot tell $\alpha_2$ from that combination of dispersions at all. (The letter $f$ follows the figure axis; the growth rate is written $f$ only inside Eqs. (1) and (4).)

**A three-band toy.** Take three bands with unit error bars and two templates, $a = (1, 2, 3)$ and $s = (2, 4, 6)$. Then $a = s/2$ exactly: the leftover is zero and $f = 1$; whatever signal $a$ could produce, $s/2$ produces identically, and the data cannot separate the two parameters. Now take $s = (2, 4, 5)$, which differs from $2a$ only on the third band. The best single weight is $c = 5/9$, the leftover is $r = (-1, -2, 2)/9$ with $r^{\mathsf{T}} r = 1/9$, and $f = 1 - (1/9)/14 = 0.992$. Whether that mismatch matters is decided by the third band's error bar. If the third band is noisy, the fit matches the first two bands almost exactly and hides the whole mismatch inside the third band's error: the leftover $\chi^2$ falls toward zero and $f \to 1$. If the third band is precise, the fit is pinned there ($c \to 3/5$) and the mismatch is pushed onto the first two bands as a visible leftover. Either way, a mismatch is only as detectable as the error bars of the bands where it must appear – which is what the $\mathsf{C}^{-1}$ weighting in Eq. (6) counts.

**One parameter against one: $f = \rho^2$.** With one $\alpha_2$ amplitude, one $\sigma_v$ and nothing else free (templates $a$ and $s$), the two-parameter Fisher matrix is built from $a^{\mathsf{T}}\mathsf{C}^{-1}a$, $a^{\mathsf{T}}\mathsf{C}^{-1}s$ and $s^{\mathsf{T}}\mathsf{C}^{-1}s$, and the correlation of its inverse is $\rho = -\,a^{\mathsf{T}}\mathsf{C}^{-1}s \,/\sqrt{(a^{\mathsf{T}}\mathsf{C}^{-1}a)(s^{\mathsf{T}}\mathsf{C}^{-1}s)}$, so $\rho^2 = f$. Mirror-image templates ($a^{\mathsf{T}}\mathsf{C}^{-1}s < 0$) give a *positive* posterior correlation: the direction the data cannot see has $\alpha_2$ and $\sigma_v$ rising together, each undoing the other's change of the spectra. The posterior correlation of the full model is a different number from $f$: it marginalizes every other parameter – cosmology, biases, the other counterterms – which can soak up part of either template, and it includes the priors, which hold parameters the data would let drift. That is why $f$ can sit at 0.99 while $|\rho|$ ranges from 0.003 to 0.90 across cuts and layouts.

**Many against many: canonical correlations.** For the full sets – six $\alpha_2$ group columns (plus tilts) against seven $\sigma_v$ columns – the squared canonical correlations generalize $\rho^2$: they are the cosines of the principal angles between the two template subspaces in the $\mathsf{C}^{-1}$ metric, the per-direction version of Eq. (6).

## 4 Results

**The templates are mirror images.** Figure 1 draws the $\alpha_2$ and $\sigma_v$ templates of three $\Delta d_{\rm com} = 62$ Mpc spectra, each scaled by its prior width. The two templates track each other band by band with opposite signs, at a few tenths of $\sigma_\ell$ per band. The bottom row is the leftover $r_\ell$ of Eq. (5): after the best combination of the seven $\sigma_v$ templates, fitted over the whole data vector, $|r_\ell|/\sigma_\ell$ stays below 0.04 per band in the BGS panel and below 0.013 in the LRG panels at the $0.2\,h$/Mpc cut, and below 0.004 everywhere at 0.075: templates of a few tenths of an error bar cancel to a few thousandths.

**The redundancy against the cut.** Over all four layouts and every cut of the scan, $f$ of the six $\alpha_2$ group columns lies between 0.958 and 1.000, with per-cut medians of 0.993–0.999; the squared canonical correlations of the six-column $\alpha_2$ set with the seven-column $\sigma_v$ set lie in the same range, 0.958–1.000. The top panel of Figure 2 draws $100\,(1 - f)$ for the least redundant group: even there, at most 4.3% of the template's information is new. The $\alpha_2$ *tilt* columns

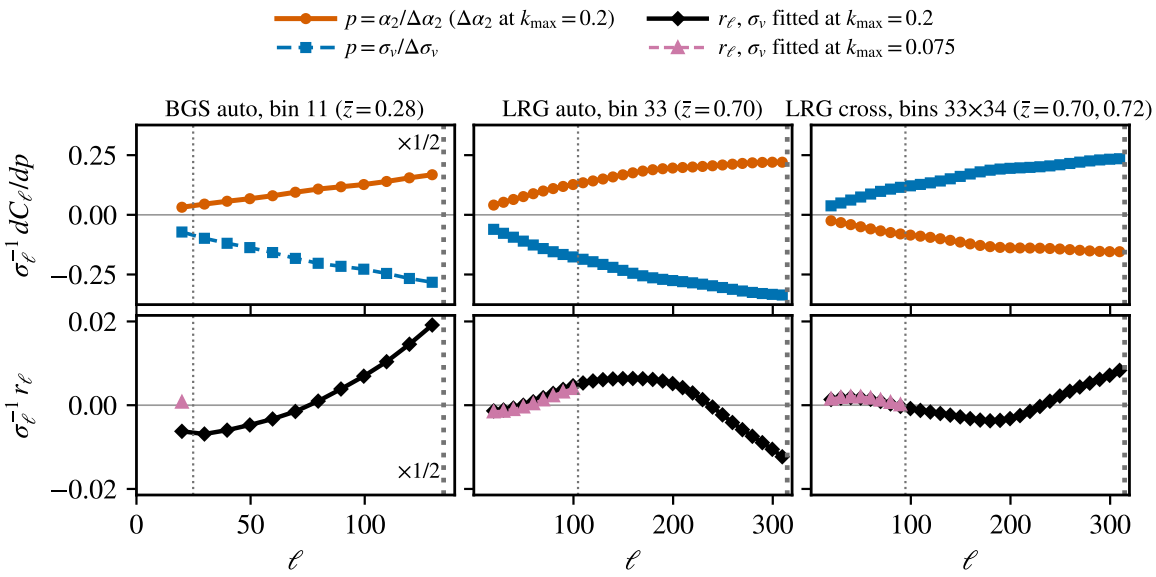



Figure 1: The $\alpha_2$ and $\sigma_v$ templates of three $\Delta d_{\rm com} = 62$ Mpc spectra – a BGS auto spectrum ($\bar{z} = 0.28$), an LRG auto spectrum ($\bar{z} = 0.70$) and that bin's cross spectrum with the next LRG bin – in units of each band's error $\sigma_\ell = \sqrt{\mathsf{C}_{ii}}$, on the bands the shipped mask keeps; the BGS column is divided by the round factor written in its panels, and each row shares one $y$ axis. Top row: the scaled templates $\sigma_\ell^{-1}\, \partial C_\ell/\partial p$, each parameter in units of its prior width: $p = \alpha_2/\Delta\alpha_2$ of the bin's group, with $\Delta\alpha_2$ the prior width at $k_{\max} = 0.2\, h/\mathrm{Mpc}$ (vermillion solid, circles), and $p = \sigma_v/\Delta\sigma_v$ of the bin's dispersion (blue dashed, squares). Bottom row: $\sigma_\ell^{-1} r_\ell$, the leftover of the $\alpha_2$ template after its best combination of the seven $\sigma_v$ templates, Eq. (5), fitted over the whole data vector in the $\mathsf{C}^{-1}$ metric on the bands a cut keeps and drawn on those bands, for $k_{\max} = 0.2$ (black, diamonds) and $0.075\, h/\mathrm{Mpc}$ (pink dashed, triangles). Dotted vertical lines: where the shown spectrum's kept bands end at 0.075 (thin) and at 0.2 (thick). $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

are much less redundant – their per-cut medians lie between 0.03 and 0.24 at every cut except the smallest cut on $\Delta d_{\rm com} = 62$ Mpc, where only three tilt columns keep any band – because a straight line in redshift across a group is a shape no combination of seven group-constant dispersions reproduces. At the template level, then, the $\alpha_2$ group values duplicate information the $\sigma_v$ already carry; the tilts are the genuinely new freedom.

**The posterior of the paper model.** The second panel of Figure 2 shows the largest $|\rho(\alpha_2, \sigma_v)|$ in the paper model's posterior. On the thin layouts it grows from 0.46 ($\Delta d_{\rm com} = 62$ Mpc) and 0.39 ($\Delta z = 0.02$) at $k_{\max} = 0.075\, h/\mathrm{Mpc}$ to 0.81 and 0.88 at 0.2, peaking at 0.84 and 0.90 at intermediate cuts; on $\Delta z = 0.05$ it stays at 0.08–0.37, and on $\Delta z = 0.2$ below 0.10. Wherever the correlation is appreciable, the pair is LRG3's $\alpha_2$ with LRG3's own dispersion, and its sign is positive, as mirror-image templates give (Section 3). The correlations stay below one because the $\sigma_v$ priors hold their end of the degenerate direction: with $\alpha_2$ free, every $\sigma_v$ posterior keeps 0.60–1.00 of its prior width, so the dispersions stay prior-dominated while $\alpha_2$

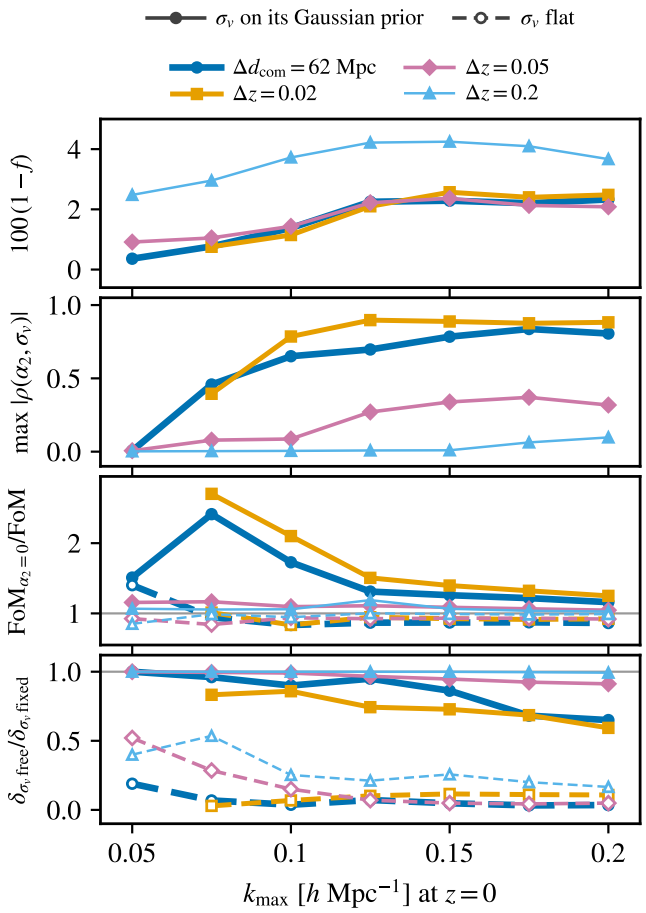



Figure 2: How far the counterterm $\alpha_2$ and the Fingers-of-God dispersions $\sigma_v$ describe the same band powers, against the scale cut, per layout (color, line width and marker; legend). Top panel: $100\,(1-f)$, the share of an $\alpha_2$ group template's information in the $\mathsf{C}^{-1}$ metric that no combination of the seven $\sigma_v$ templates reproduces, for the group where that share is largest; this panel is built from the templates and the covariance alone and does not depend on any prior. Second panel: the largest $|\rho|$ between an $\alpha_2$ parameter and a $\sigma_v$ in the posterior of the paper model, model B + tilts on the pessimistic linear-bias prior. Third panel: $\mathrm{FoM}_{\alpha_2=0}/\mathrm{FoM}$, the $\mathrm{FoM}(w_0, w_a)$ with $\alpha_2$ and its tilts fixed at zero over the paper model's, with $\sigma_v$ on its Gaussian priors (solid, filled markers) or flat (dashed, open markers). Bottom panel: $\delta_{\sigma_v\,\mathrm{free}}/\delta_{\sigma_v\,\mathrm{fixed}}$, with $\delta$ the $(w_0, w_a)$ shift of model A + tilts, which has no $\alpha_2$, fitted to a truth with $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group: the shift with $\sigma_v$ free over the shift with $\sigma_v$ fixed at its fiducial, with the same two line styles. Gray: one. $\Delta z = 0.02$ keeps no band at $0.05\,h/\mathrm{Mpc}$. $k_{\rm max}$ quoted at $z = 0$; each bin is cut at $k_{\rm max}/D(\bar{z}_i)$.

takes the $k_\parallel^2$ information. With $\alpha_2$ and its tilts fixed, the data begin to measure the dispersions: on the thin layouts the best-measured $\sigma_v$ tightens to 0.25–0.74 of its prior width at cuts of 0.075 and above.

**What $\alpha_2$ costs.** The third panel of Figure 2 draws $\mathrm{FoM}_{\alpha_2=0}/\mathrm{FoM}$: the paper model with $\alpha_2$ and its tilts fixed at zero, over the paper model. On the thin layouts the fixed fit has 2.41 and 2.70 times the $\mathrm{FoM}(w_0, w_a)$ at $k_{\max} = 0.075\,h/\mathrm{Mpc}$, 1.73 and 2.10 at 0.1, and 1.16 and 1.25 at 0.2; $\mathrm{FoM}(\Omega_m, 10^9 A_s)$ gains similar factors (2.27 and 2.19 at 0.075, 1.18 and 1.25 at 0.2). On $\Delta z = 0.05$ the gain is 1.05–1.17, and on $\Delta z = 0.2$ at most 1.07 at every cut except 0.125, where it reaches 1.19. The dashed curves isolate where the gain comes from: with the $\sigma_v$ priors removed (flat), fixing $\alpha_2$ returns 0.83–1.01 of the paper model's FoM at every cut from 0.075 on – no gain at all. Fixing $\alpha_2$ therefore does not add information of its own; it hands the $k_\parallel^2$ direction to the dispersions, and the constraint the fixed fit gains is exactly the constraint the $\sigma_v$ priors supply.

**What dropping $\alpha_2$ risks.** The absorption test fits model A + tilts, which has no $\alpha_2$, to a truth with $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group – the DESI prior width at $k_{\max} = 0.2\,h/\mathrm{Mpc}$. If $\sigma_v$ could stand in for $\alpha_2$ in practice, freeing it would absorb the shift. It barely does: with $\sigma_v$ on its Gaussian priors, the $(w_0, w_a)$ shift falls from the fixed-$\sigma_v$ value by 4% at 0.075 ($0.335 \to 0.322$ ellipse units) and 35% at 0.2 ($1.65 \to 1.07$) on $\Delta d_{\mathrm{com}} = 62$ Mpc, and by 17% ($0.305 \to 0.254$) and 41% ($2.60 \to 1.54$) on $\Delta z = 0.02$; on $\Delta z = 0.05$ by at most 9%, and on $\Delta z = 0.2$ the shift never exceeds $0.016\sigma$ to begin with. With flat $\sigma_v$ priors the same shifts, at cuts of 0.075 and above, fall by 88–97% (bottom panel of Figure 2): the dispersions *can* mimic the counterterm, but only by moving far outside their priors. Fitting the $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth with the $\sigma_v$ templates alone makes this concrete: on $\Delta d_{\mathrm{com}} = 62$ Mpc at 0.2 the truth carries $\Delta\chi^2 = 162$, of which free dispersions leave 1.6 – but the mimicking shift is $-2.5$ prior widths ($-75$ km/s, always the ELG2 dispersion, whose prior is tightest), and with the Gaussian priors the total left, data residual plus prior penalty, is 14. Across the thin layouts at cuts of 0.075 and above, the mimicking shift is 1.9–2.6 prior widths (57–78 km/s). The shifts scale linearly with the truth: a truth at one model B prior width ($0.5/k_{\max}^2$, $88.9\,h^{-2}\,\mathrm{Mpc}^2$ at 0.075) gives 7.1 times the $12.5\,h^{-2}\,\mathrm{Mpc}^2$ shifts there.

## 5 How wide must the $\sigma_v$ priors be?

**The knob.** The absorption test leaves a natural follow-up question: if $\sigma_v$ keeps a Gaussian prior, how wide must that prior be for the dispersions to absorb a DESI-size $\alpha_2$? The two preceding results frame the answer before any scan. Mimicking $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ costs the dispersions 1.9–2.6 prior widths, so widening the priors by a factor of a few makes that move cheap and the shift disappears. But the FoM gain of dropping $\alpha_2$ comes entirely from those same priors, so the same widening surrenders the gain. A single knob – a common factor $s$ on all seven prior widths of Table 1 – therefore slides the fit between two ends: at $s = 1$ the full gain and almost no absorption, with flat priors the full absorption and no gain. The scan locates where between the two ends each layout and cut sits.

**The scan.** The fit is the one of the absorption test – model A + tilts on the pessimistic linear-bias prior, which carries $\alpha_0$ but no $\alpha_2$: the $\alpha_0$-only setup of Section 2 – fitted to the truth $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group, with all seven $\sigma_v$ prior widths multiplied by $s \in \{1, 1.5, 2, 3, 5, 10\}$ and flat. Each $s$ is scored by $\delta_{\max}$, the largest of the $(w_0, w_a)$ shift in its ellipse and the $\Omega_m$ and $A_s$ shifts in their own $\sigma$, and by $\mathrm{FoM}_{\alpha_0\,\mathrm{only}}/\mathrm{FoM}$, that fit's $\mathrm{FoM}(w_0, w_a)$ over the paper model's at $s = 1$. The passing $s$ is the smallest $s$ with every shift below $\sigma/3$, interpolated linearly in $\log s$ between the scanned values. Figure 3 draws the scan; every cut, the per-shift values and the one-sample variants below are in that folder's `data.json`.

**At $k_{\max} = 0.075\,h/\mathrm{Mpc}$ the default priors almost suffice.** $\Delta d_{\mathrm{com}} = 62$ Mpc passes at $s = 1.84$, keeping 2.13 times the paper model's FoM; $\Delta z = 0.02$ passes at $s = 1$, keeping 2.77. $\Delta z = 0.05$ passes at $s = 1$ at every cut up to 0.15 (FoM ratios 1.12–1.23) and needs $s \approx 2.9$ at 0.175–0.2, where the ratio is 1.02–1.03; $\Delta z = 0.2$ passes at $s = 1$ everywhere (1.05–1.25).

**At 0.1–0.2 on the thin layouts the two options converge.** There the passing factor is $s = 3.56$–$3.83$ on $\Delta d_{\mathrm{com}} = 62$ Mpc and 4.83–5.99 on $\Delta z = 0.02$: ELG2's 30 km/s prior becomes about 110–180 km/s, a third to more than half of its fiducial dispersion. At those widths the FoM ratio is 0.96–1.22: the widened $\alpha_0$-only fit and the paper model constrain dark energy essentially equally, and the gain that motivated dropping $\alpha_2$ is gone.

**One sample's dispersions alone do not absorb it.** The truth holds $\alpha_2$ in every group, so the mimicking must too. Widening only the three LRG dispersions fails wherever $s = 1$ does not already pass: on $\Delta d_{\mathrm{com}} = 62$ Mpc at 0.2, $\delta_{\max}$ falls only from 1.16 to 1.12 even with the LRG priors flat. Widening only the two ELG dispersions passes on $\Delta d_{\mathrm{com}} = 62$ Mpc at 0.075 ($s = 2.94$, ratio 2.24) and on $\Delta z = 0.05$ at 0.175–0.2, but fails on the thin layouts at 0.1 and above. Absorbing $\alpha_2$ needs all the dispersions widened together.

## 6 The pair list at the edges of the $\sigma_v$ prior

**The rule.** The data vector measures a fixed list of cross spectra, chosen by the Fingers-of-God overlap rule (`scripts/make_desi_pairs.py`; `notes/cross_correlations` derives it). With the normalized overlap of two bins' blurred $n(z)$,

$$O_{ij} \;=\; \frac{\int dz\, n_i(z)\, n_j(z)}{\sqrt{\int dz\, n_i^2(z)\, \int dz\, n_j^2(z)}}, \tag{7}$$

1 for two identical shapes, 0 for disjoint ones and, at high $\ell$, close to the correlation coefficient of the two bins' spectra, the list holds DESI's measured pairs plus every pair with $O_{ij} \geq 10^{-6}$ at the fiducial $\sigma_v$. The list and the covariance are built once, at the fiducial dispersions; a sampled $\sigma_v$ changes every kernel but never the list.

**Raising $\sigma_v$ pushes unlisted pairs toward the threshold.** A larger $\sigma_v$ widens every kernel, so every overlap grows – the listed ones, and those of the pairs outside the list, bins of one sample two apart, whose blurred kernels are separated by a full bin. Figure 4 follows the largest unlisted overlap as every dispersion group moves to $\sigma_v^{\mathrm{fid}} + k\,s\,\Delta\sigma_v$, with $k$ the offset in

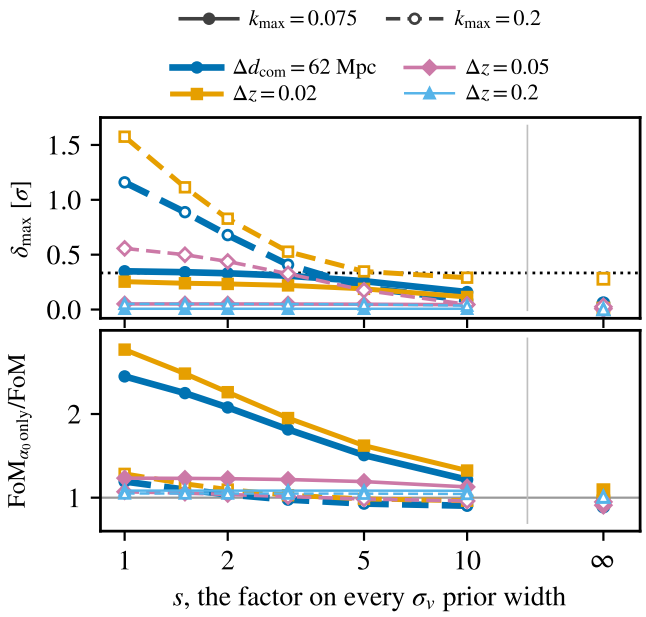



Figure 3: The $\sigma_v$ prior-width scan: the $\alpha_0$-only fit (model A + tilts on the pessimistic linear-bias prior) fitted to a truth with $\alpha_2 = 12.5\,h^{-2}\,\mathrm{Mpc}^2$ in every group, against $s$, the common factor on all seven $\sigma_v$ prior widths (log axis; the flat-$\sigma_v$ points sit at the right edge, past the gap), per layout (color, line width and marker; legend) at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ (solid, filled markers) and 0.2 (dashed, open markers). Top panel: $\delta_{\max}$, the largest of the $(w_0, w_a)$ shift in its ellipse and the $\Omega_m$ and $A_s$ shifts in their own $\sigma$; dotted: $\sigma/3$. Bottom panel: $\mathrm{FoM}_{\alpha_0\,\mathrm{only}}/\mathrm{FoM}$, that fit's $\mathrm{FoM}(w_0, w_a)$ over the paper model's, model B + tilts with $\alpha_2$ free and $\sigma_v$ at $s = 1$; gray: one. $k_{\max}$ quoted at $z = 0$; each bin is cut at $k_{\max}/D(\bar{z}_i)$.

prior widths and $s$ the width factor of Section 5. At the fiducial it is zero on every layout: no unlisted kernels meet. At $k = 1$, 2 and 3 with $s = 1$ it reaches $1.7 \times 10^{-10}$, $7.6 \times 10^{-7}$ and $2.9 \times 10^{-5}$ on $\Delta d_{\rm com} = 62$ Mpc (a pair of BGS bins two apart), and $8.6 \times 10^{-10}$, $5.0 \times 10^{-8}$ and $7.9 \times 10^{-7}$ on $\Delta z = 0.02$ (ELG2, then LRG3 bins two apart), while $\Delta z = 0.05$ and 0.2 stay at zero. The threshold is crossed at $k = 2.06$ on $\Delta d_{\rm com} = 62$ Mpc, 3.11 on $\Delta z = 0.02$, 12.5 on $\Delta z = 0.05$ and nowhere on $\Delta z = 0.2$. With the priors widened by $s = 4$, the order Section 5 requires at 0.1–0.2, the thin layouts cross already at $k \approx 0.5$–0.8.

**The paper model stays inside the list.** At the $+2\sigma$ edge of the current priors the largest unlisted overlap is $7.6 \times 10^{-7}$ on $\Delta d_{\rm com} = 62$ Mpc – a factor 1.3 below the threshold – and $5.0 \times 10^{-8}$ on $\Delta z = 0.02$; at +2 posterior widths of the paper model it is the same (at 0.075, where the posterior equals the prior) or smaller (at 0.2). A chain over the current priors therefore keeps every pair it needs in the list; $\Delta d_{\rm com} = 62$ Mpc crosses only just beyond, at +2.1 prior widths.

**What $\alpha_2$ asks of $\sigma_v$.** Only a *negative* $\alpha_2$ drives the dispersions up: a positive $\alpha_2$ looks like a smaller $\sigma_v$ (Section 4), so its mirror image raises them. In the linear response of the fit without $\alpha_2$, one unit of $\alpha_2$ moves the most responsive dispersion by 0.1–2.2 km/s with the priors on (the pull grows with the cut) and by 5.4–7.2 km/s with flat priors (on every layout but $\Delta z = 0.2$). The $|\alpha_2|$ at which an unlisted pair first reaches $10^{-6}$ is, with flat $\sigma_v$: 16–19 $h^{-2}\,\mathrm{Mpc}^2$ on $\Delta z = 0.02$ – 1.3–1.5 times the DESI prior width $12.5\,h^{-2}\,\mathrm{Mpc}^2$ – through a pair of ELG2 bins two apart; 53–59 on $\Delta d_{\rm com} = 62$ Mpc (ELG1); 105–133 on $\Delta z = 0.05$; and 394–462 on $\Delta z = 0.2$. With the current priors it is at least $68\,h^{-2}\,\mathrm{Mpc}^2$ at $k_{\rm max} = 0.2\,h/\mathrm{Mpc}$ and at least $1000\,h^{-2}\,\mathrm{Mpc}^2$ at 0.075.

**The caution for chains.** The pair list and the covariance are built once, at the fiducial $\sigma_v$. The range of $\alpha_2$ a chain treats as reasonable therefore bounds how far $\sigma_v$ travels when it has to absorb $\alpha_2$, and the widened or flat $\sigma_v$ of Section 5 is exactly the regime where the list stops being complete: dispersions high enough to mimic a DESI-size negative $\alpha_2$ also raise unlisted pairs – bins two apart – past the threshold, so more cross-correlations would have to enter the data vector. Either the chain bounds $\sigma_v$ (or $\alpha_2$) within the ranges above, or the list and covariance are rebuilt at the prior edge.

**The listed pairs' own overlaps.** The complementary test – keeping only the listed pairs whose overlap reaches a floor $O_{\rm min}$, DESI-measured pairs included – drops, for every floor up to $10^{-3}$, one pair: the BGS–LRG1 boundary pair of $\Delta d_{\rm com} = 62$ Mpc ($O = 3.4 \times 10^{-7}$), at a cost of 0.1% of FoM$(w_0, w_a)$ at $k_{\rm max} = 0.2\,h/\mathrm{Mpc}$ and nothing elsewhere, so none of this note's conclusions rests on a low-overlap pair. The study and its limits are in `notes/cross_correlations`, section "How much do the low-overlap pairs carry?".

# 7 What this means

At the template level, $\alpha_2$ and the sampled $\sigma_v$ carry the same $k_\parallel^2$ information: in that sense the model double counts one physical effect, and the $\alpha_2$ group values add almost no template direction the dispersions do not already span. Only the $\alpha_2$ tilts are new.

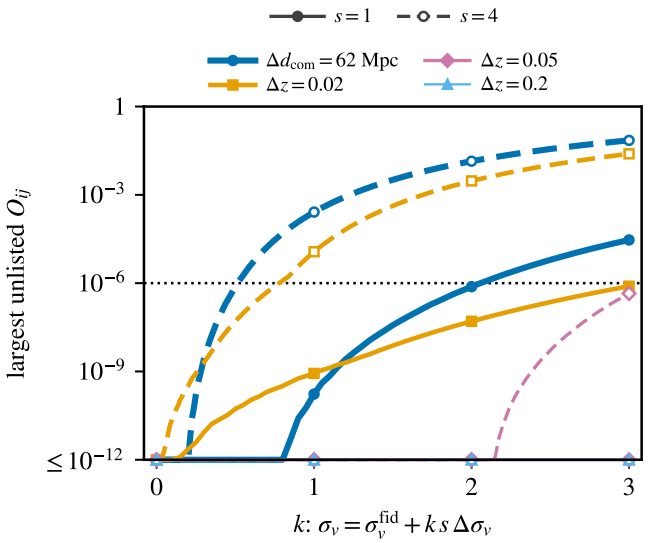



Figure 4: How far the pair rule holds away from the fiducial dispersions: the largest overlap $O_{ij}$ of Eq. (7) over the pairs *outside* each layout's pair list, against the offset $k$ of every $\sigma_v$ group from its fiducial, in units of its prior width times the width factor $s$ ($\sigma_v = \sigma_v^{\mathrm{fid}} + k\, s\, \Delta\sigma_v$), per layout (color, line width and marker; legend). Solid, filled markers: $s = 1$, the current priors; dashed, open: $s = 4$, the order of the widths that let $\sigma_v$ absorb a DESI-size $\alpha_2$; markers at $k = 0$, 1, 2, 3. Dotted: the rule's threshold $O_{ij} = 10^{-6}$, above which a pair belongs in the data vector. Overlaps below the axis floor, zero included (kernels that do not meet), are drawn on the floor.

In the posterior, the Gaussian $\sigma_v$ priors break the degeneracy. They are tight enough (10–40% of the fiducial dispersions) that the dispersions cannot wander along the degenerate direction, so the joint fit stays well conditioned at every cut, with the largest $\alpha_2$–$\sigma_v$ correlation reaching 0.9 only at the largest cuts on the thin layouts.

Keeping both parameters has a definite division of labor: $\alpha_2$, with its wide prior, takes the $k_\parallel^2$ information, and the dispersions fall back to their priors (0.60–1.00 of the prior width with $\alpha_2$ free). The data then constrain nonlinear velocities only as far as the $\alpha_2$ prior allows, and the $\sigma_v$ priors contribute almost nothing to the FoM.

Dropping $\alpha_2$ gains a factor of 2.4–2.7 in FoM$(w_0, w_a)$ at $k_{\rm max} = 0.075\,h/$Mpc and 1.16–1.25 at 0.2 on the thin layouts – a gain that is exactly as good as the $\sigma_v$ priors, since it vanishes when those priors are removed. The same move carries the bias risk of the absorption test: if a DESI-prior-sized $\alpha_2$ is present in the truth, a fit without $\alpha_2$ is shifted by up to 0.34 ellipse units at 0.075 and 1.6–2.6 at 0.2 on the thin layouts, and the dispersions on their priors absorb at most 4–41% of it. The wide layouts barely care either way: their gains are at most $\sim$20% and their shifts at most $0.43\sigma$ ($\Delta z = 0.05$ at 0.2).

These are the trade-offs as measured; whether the forecast keeps $\alpha_2$, drops it in favor of the dispersions, or moves to a setup with $\alpha_0$ only is not decided in this note (appendix "Open issues").

# A Method and checks

**Machinery.** Everything runs on the model B notebook's final section (Section 1 names the file): the final-tier derivative caches of that notebook (five-point stencils; the $\alpha_2$ columns are exact one-difference templates, since the term is linear in its amplitude), the Gaussian band-power covariance restricted to the bands each cut keeps, and the notebook's Fisher systems $F = \mathsf{D}^\mathsf{T}\mathsf{C}^{-1}\mathsf{D} + F_{\rm priors}$, with $\mathsf{D}$ the derivative matrix and $F_{\rm priors}$ the Gaussian priors of the paper model (the pessimistic linear-bias prior, the one-loop offset and counterterm priors, and the $\sigma_v$ priors of Table 1). Posterior correlations are read off $F^{-1}$; the fixed-$\alpha_2$ and flat-$\sigma_v$ variants remove the $\alpha_2$ columns or the $\sigma_v$ prior rows and rebuild $F$.

**The absorption test is a linear response.** The shift of model A + tilts under an $\alpha_2$ truth is the Fisher linear response $\delta p = F^{-1}\mathsf{D}^\mathsf{T}\mathsf{C}^{-1}\,\delta m$, with $\delta m$ the truth's change of the data vector, scored in the fit's marginalized $(w_0, w_a)$ ellipse ($\sigma$ for one parameter) – not a refit. Shifts of one to two ellipse units extrapolate that linear response accordingly.

**Checks.** The analysis reproduces, to the printed precision (0.0 relative difference), the paper model's FoM$(w_0, w_a)$ and FoM$(\Omega_m, 10^9 A_s)$ of the model B notebook's `fom_kmax_models` figure data at every cut, and the model A + tilts $\alpha_2$-truth shifts of the notebook's break-test scores. The $\sigma_v$-template fits of the $\alpha_2 = 12.5\,h^{-2}\,{\rm Mpc}^2$ truth satisfy the linear-algebra identities of Eq. (6) by construction; the notebook's own self-checks on the $\alpha_2$ columns (the data vector returns bitwise after the template evaluation; forward and backward differences agree) are listed with that notebook.

# B Open issues

- **Keep or drop $\alpha_2$.** The trade is measured in this note – a factor 2.4–2.7 of FoM$(w_0, w_a)$ at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ on the thin layouts against the absorption shifts if a DESI-sized $\alpha_2$ is real – but the decision is not made here. The prior-width scan sharpens what it is: in effect a choice between keeping $\alpha_2$ and an $\alpha_0$-only model whose $\sigma_v$ prior widths are multiplied by the passing $s$ of Section 5, and the two converge in FoM at $k_{\max} \geq 0.1$ on the thin layouts, so the cut at which the forecast is quoted matters to the choice.

- **The basis of the $\sigma_v$ priors.** The gain from dropping $\alpha_2$ is exactly as strong as the $\sigma_v$ priors, so it inherits their basis: Gaussians five times as wide as the per-bin error bars of the source forecast's measurement on simulated galaxies, truncated at zero (`notes/cross_correlations`). How defensible that width is for the real samples, and whether a Gaussian blur is the right shape for the small-scale velocity distribution, are open.

- **No redshift-space one-loop terms.** The sampled $\sigma_v$ and $\alpha_2$ are the model's only handles on nonlinear velocities; a velocity effect beyond a $k_\parallel^2$ correction is outside both.

- **Linear response.** The largest absorption-test shifts (1.5–2.6 ellipse units) are linear extrapolations; a refit could move them.

- **The pair list at the $\sigma_v$ prior edge.** Building the pair list at the prior edge adds the pairs of bins two apart (Section 6); it needs new spectra, the four covariances and new derivative caches, and is queued as an overnight study.

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